Damien Rouchouse

Master's student in mathematics and machine learning – PhD applicant
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Education

École normale supérieure Paris-Saclay — Gif-sur-Yvette, France
M.Sc. Mathematics, Vision and Learning (MVA) — 2025–2026

- Relevant coursework: optimal transport, convex optimization, computational statistics, geometric data analysis, graph machine learning, random matrix theory, inverse problems, generative models, stochastic calculus and robotics.

Mines Paris – PSL — Paris, France
Diplôme d'ingénieur (M.Sc. equivalent), Mathematics and Computer Science — 2022–2026

- Relevant coursework: measure theory, probability, statistics, machine learning, generative models, functional analysis and signal processing.

Lycée Hoche — Versailles, France
CPGE MPSI–MP – intensive mathematics and physics program* — 2020–2022

- TIPE: Radon transform for tomography.

Research Experience

École normale supérieure – Department of Mathematics and Applications (DMA) — Paris, France
Research Intern – Weak error analysis for discrete diffusion samplers — 2026–present
Supervisors: Julie Delon, Rémi Gribonval and Gabriel Peyré

- Derived the leading weak-error term of the matrix Euler discretization for Bayesian time reversals of finite-state continuous-time Markov chains.
- Obtained a spectral representation showing how the bias depends on the corruption generator, noise schedule, data distribution, and test observable.
- Analyzed the interaction between terminal mismatch, reverse-rate perturbations, and Euler bias on a two-state graph.

Inria — Lyon, France / London, UK
Research Intern – PAC-Bayesian generalization bounds — 2025 (4 months)
Supervisors: Antoine Gonon, Rémi Gribonval, and Benjamin Guedj

- Studied neuron-wise rescaling symmetries of ReLU networks and their effect on PAC-Bayes complexity terms.
- Formulated PAC-Bayes bounds in an invariant lifted representation and analyzed their guarantees through data processing.
- Implemented KL-based optimization procedures and evaluated them on neural-network experiments using PyTorch Lightning and Weights & Biases.

Harvard SEAS – Bertoldi Group — Boston, USA
Research Intern – Programmable origami metamaterials — 2024 (5 months)

- Modeled compatibility constraints for programmable origami patterns.
- Studied multistable transitions using Abaqus simulations and experimental validation on macro- and microscale prototypes.
- Implemented a Python pipeline generating DXF fabrication files from geometric design parameters.

Preprint

- **D. Rouchouse**, A. Gonon, R. Gribonval, and B. Guedj. "Non-Vacuous Generalization Bounds: Can Rescaling Invariances Help?" arXiv:2509.26149, 2025.

Additional Experience

Scortex – TRIGO Group — Paris, France
Computer Vision Intern — 2024–2025 (6 months)

- Adapted diffusion- and distillation-based computer vision methods for high-speed industrial anomaly detection.
- Benchmarked and deployed selected models under real-time constraints using PyTorch and MLflow.

French Ministry of Education — Versailles, France
Oral Examiner in Mathematics — 2023–2024 (6 months)

- Prepared and assessed oral mathematics examinations for students preparing engineering-school entrance examinations.

Technical Skills

Programming: Python (PyTorch, PyTorch Lightning, NumPy, scikit-learn), working knowledge of C/C++.
Tools: Git, Linux, Docker, LaTeX, MLflow, Weights & Biases.
Languages: French (native), English (TOEFL iBT: 110/120).